

American Academy of Pediatrics

DEDICATED TO THE HEALTH OF ALL CHILDREN®



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Sethuraman Panchanathan, PhD

Director

National Science Foundation

2415 Eisenhower Ave

Alexandria, VA 22314

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Dear Dr. Panchanathan:

On behalf of the American Academy of Pediatrics (AAP), a non-profit professional organization of more than 67,000 primary care pediatricians, pediatric medical subspecialists, and pediatric surgical specialists dedicated to the health, safety, and well-being of all infants, children, adolescents, and young adults, I write to provide comments in response to the request for information (RFI) from the Office of Science and Technology Policy (OSTP) regarding the development of an Artificial Intelligence (AI) Action Plan.

The AAP appreciates the opportunity to provide comments on this RFI, and the following provides the Academy's input on several areas outlined in the RFI for consideration to inform the development of the AI Action Plan.

Cybersecurity

In order to strengthen the AI marketplace, the AAP believes it is essential that cybersecurity, data intelligence practices, and classification of data are placed under specifically outlined security classifications that are easy to follow as necessary measures for assessing benefits and trade-offs. Transparent policies and transparency regarding the development and use of the technology that does not prevent assessment, comparison, and versioning would be helpful, particularly in knowing the extent of training data. Higher levels of security classification requiring in-house infrastructure with robust cybersecurity infrastructure that has minimum requirements for ages of systems, operating systems, cloud security etc. would need careful assessment in any area to be deployed. Lower levels of security may be considered for contracted development, licensing of AI software, and similar use cases, again with specific requirements outlined to any contracting entity around designs, proof of concept, beta testing, and surveillance of any systems with agreement to oversight in place. Further, independent oversight is essential to ensuring objective assessments of performance and security testing.

Model Development

It is essential that the design and algorithms of AI models take into account who designs the algorithms and what information is utilized to train the model. The AAP recommends specific guidelines around the process of selecting authors, reviewers, and beta testers from a variety of backgrounds, including age and sex, to minimize biases and ensure fairness.

Application and use

AI models used in pediatric health care must be trained on appropriately representative pediatric data, given the physiological, developmental, and social differences between children and adults. Special considerations should be made for developmental stage-specific AI tools, ensuring that AI recommendations account for age-related variations in diagnosis and treatment. We also urge caution when deciding when and how to use AI in health care settings; there are some practice areas in which the use of AI, such as predictive technology, could have adverse outcomes.¹ AI should be used to enhance, not replace, pediatric clinical decision-making, particularly in high-risk areas like neonatology, child abuse assessments, and mental health.

Additionally, in the early stages of incorporating AI systems, it is vital to maintain a focus on function/decision making within the algorithms. Initially, the systems should be trained to learn very specific functions but not perform functions or make decisions that affect people directly. Furthermore, a transparent risk/benefit/potential harm analysis should be conducted, including what underlying data sets were used to inform the technology and whether this technology should be only applied/restricted to certain populations. Administrative processes that are straightforward, requiring some specific information gathering with finite data sets and decision tools, should be performed initially. For example, within the health care setting, AI may be able to help with a prior authorization document for obtaining authorization from insurance entities for an alternative medication or piece of equipment. As mentioned previously, beta testing this infrastructure is critical and consideration to parallel their function with a current state to evaluate accuracy prior to any real-world deployment.

Further, transparency is important when utilizing AI tools. As an example, AI tools are often used for processing payment claims and reading notes for review/appeal, and it would be helpful for both patients and providers to be aware that AI tools were utilized for these purposes. The AAP recommends a tag or acknowledgement such as "this patient care note was generated using ambient listening and AI" as well as the name of the company whose product is being used.

Education and workforce

The AAP strongly recommends that AI-based educational tools be rigorously evaluated to prevent potential developmental harm, particularly in early childhood learning and socialization. Transparency is crucial in AI-powered digital learning tools, ensuring parents and educators understand how algorithms influence children's educational experiences. Additionally, AI-driven mental health applications for children (e.g., chatbots, emotion detection tools, etc.) should have clear ethical guidelines, with an emphasis on preventing unintended psychological effects.

Regarding workforce education, pediatricians and child health professionals should receive dedicated training in AI literacy, including how AI systems generate recommendations, potential biases, and limitations in pediatric contexts. Within the AI Action Plan, the AAP recommends including support for medical education programs that integrate AI ethics and practical AI use cases in pediatrics. Federal agencies should work with medical societies, including the AAP, through the standard public notice and comment process under the Administrative Procedure Act, as well as other advisory bodies such as those created by the Federal Advisory

Committee Act, to create standardized AI competency guidelines for pediatric healthcare professionals in a transparent and informed manner.

Explainability and assurance of AI model outputs

Maintaining clear and consistent definitions and expectations surrounding AI/machine learning (ML) for companies is important. Updates to the regulatory frameworks that provide guidance regarding needs for explainability and interpretability rather than simple monitoring of outcomes will help companies understand what is needed for particular use cases. AI tools used in pediatrics should have clear explainability requirements to ensure that clinicians and families understand AI-generated recommendations. Parents and guardians should have access to AI decision-making rationales when AI is used to support or automate healthcare-related decisions affecting their children.

Similar to clinical decision support or references in electronic health records (EHRs) such as growth charts derived from the Centers for Disease Control and Prevention (CDC) or World Health Organization (WHO) growth curves, it would be helpful to include a way to link to references to increase assurance of AI model outputs. This would allow users to use the information provided by AI systems or services for "assistance" but if it is not consistent with what the professional knows, they can access the reference and have a way to confirm that either the technology is wrong or that they had incomplete knowledge.

Data privacy and security throughout the lifecycle of AI system

Data privacy and security is essential for the successful adoption of AI within clinical settings. Children's data is particularly sensitive and has long-term implications for identity protection. As such, AI policies should include stricter protections for pediatric data than for adult data. AI models that interact with pediatric patient data should comply with both the Health Insurance Portability and Accountability Act (HIPAA) and the Children's Online Privacy Protection Act (COPPA) to ensure that privacy measures are in place. Further, the AAP recommends that AI systems handling pediatric data are required to incorporate privacy-preserving techniques such as differential privacy and secure multi-party computation. While the protections afforded by HIPAA and COPPA are essential to protecting children, there are at present no broad-based, nationwide data privacy protections in place to protect all children and adolescents, a reality which has led to widespread data collection from young people across their devices and accounts. This has significant implications for the development of AI, which depends on the processing of vast quantities of data and may present significant risks to user privacy and AI model performance, particularly as AI products increasingly blur the line between everyday uses and clinical care. We believe a robust data privacy framework to protect all young people is essential to the successful development of AI.

Overall, the HIPAA framework is reasonable for safeguarding patient privacy in clinical settings, but the HHS Office of Civil Rights (OCR) will need to further clarify and equip the existing HIPAA framework to safeguard patient privacy with regards to AI. Using Microsoft Azure as an example, to provide HIPAA-compliant services, they must assume that HIPAA violations could occur in any patient inputs. Though unlikely, there are fringe cases where an algorithm could accept user input as training data and then inadvertently share it in response to a separate patient request, leading to a HIPAA infraction. While these cases would be somewhat unusual, they represent a risk with applying any large language model (LLM). At some point, even with significant human reinforcement training, algorithms may behave unpredictably in unusual or fringe cases. HIPAA would likely qualify these unusual cases as violations in the current form, but as stated previously, further clarification is needed.

Risks

AI systems are designed to learn from human interactions and have limited ability to filter information with human context around sociocultural infrastructure and abstract human interactions. As such, it is necessary to assume that these systems will carry implicit bias and pose a significant risk of major errors in learning and returning responses as a result. The AAP recommends intentional mitigation efforts at several levels to ensure AI systems or services advance equitable outcomes and mitigate risks.

These systems also pose significant risks to the health and development of young people. Anecdotal experience suggests that AI-powered image generation tools have allowed for the creation of falsified, explicit images, often known as Deepfakes, that can leave young people feeling violated and humiliated. In one case, reporting has indicated that students at one high school created Deepfakes of a classmate based on photos found online and circulated those images to other students. This deeply concerning example highlights the perils of developing AI without accounting for the developmental age of young people. The widespread commercialization of these powerful technologies will unfortunately lead young people of all ages to test the limits of the technology, beyond the use cases likely envisioned, with the potential to lead to harmful outcomes. A framework for the development of AI must account for the developmental stage of young people as these tools become a growing part of young people's lives.

Regulation and governance

Metrics beyond the performance of AI on leaderboards or tasks developed by the computer science community need to be created, particularly those which seek to examine the linguistic properties of text beyond syntactic or grammatical measures. AI should be treated like any other technology, and robust measurement of performance beyond project objectives (e.g. sustainability, validity, reliability) should be identified. Additionally, ensuring that there is a clear understanding of the data definitions of performance and clear goals of performance and performance surveillance is critical. The AAP also recommends that mechanisms are put in place to collect and review unintended consequences for those who perceive negative impacts of this technology and to ensure that there is accountability for any harm caused by AI-enabled tools or services.

Additionally, federal agencies should collaborate with the AAP and other experts to develop pediatric-specific AI standards, including safety benchmarks and performance metrics. The AAP recommends AI regulation that incorporates a pediatric risk assessment framework, ensuring that AI used in pediatric care does not inadvertently reinforce biases, compromise safety, or lead to inappropriate decision-making. For example, pediatric-specific AI governance frameworks should address concerns related to AI-assisted child welfare decision-making, ensuring that AI does not exacerbate disparities in social services or foster care.

Procurement

Any procurement should be driven by a willingness to use generative AI appropriately, rather than explicitly altering the technology. Particularly in the context of removing images and videos from a platform, enhancing the use of barcoding or digital watermarking technology can potentially serve as a first line of defense, but it is possible that the adaptation of commercial systems will result in such technology being subverted or circumvented.ⁱⁱ Ultimately, it will be critical to ensure that the technology's use is tracked and managed properly.

The Academy recommends establishing standard pre-procurement specific questions around products and systems to help determine whether they incorporate any type of AI, as these would be important for evaluation. An initial definition of AI would be important as a starting point to develop a tool that can be leveraged to vendors around any systems being offered. This would require AI expertise, but a specific assessment tool can be developed as a standard process for evaluation of any system for procurement.

Thank you again for the opportunity to provide comments on this request for information. The Academy would welcome the opportunity to work with OSTP as the AI Action Plan is developed and implemented. If we can be of further assistance, please contact Patrick Johnson in our Washington, DC office at pjohnson@aap.org.

Sincerely,

Susan J. Kressly, MD, FAAP
President

SJK/ncp

ⁱ See, "[Does artificial intelligence discriminate in child neglect case assessments?](#)", ABA Journal, December 1, 2023.

ⁱⁱ Nguyen A, Yosinski J, Clune J. Deep Neural Networks are Easily Fooled: High Confidence Predictions for Unrecognizable Images. In Computer Vision and Pattern Recognition (CVPR '15), IEEE, 2015.